TEXORA · LECTURE SHEET

Finishing Machinery

Finishing machines give fabric its final dimensions, surface and hand. The stenter is their centre — dryer, width-setter and chemical curing range in one — supported by compactors, sanforizers, calenders and surface machines.

WP-3207 · ADVANCED MACHINERIES AND RELEVANT CALCULATIONS · TOPIC 9 · SPECIALIZATION — WET PROCESSING

1. The finishing line and its tasks

After dyeing or printing and washing, fabric is wet, distorted and dimensionally unstable. Finishing must: extract water and dry; restore and set width and length dimensions; apply and fix chemical finishes (Topic 4); and create the final surface (soft, glazed, raised, or matte). The machines divide into **wet-processing units** (padder, vacuum extractor, hydro-extractor), the **stenter**, **dimensional machines** (compactor, sanforizer), and **surface machines** (calender, raising, shearing, sueding, decatising).

2. The stenter — the central finishing machine

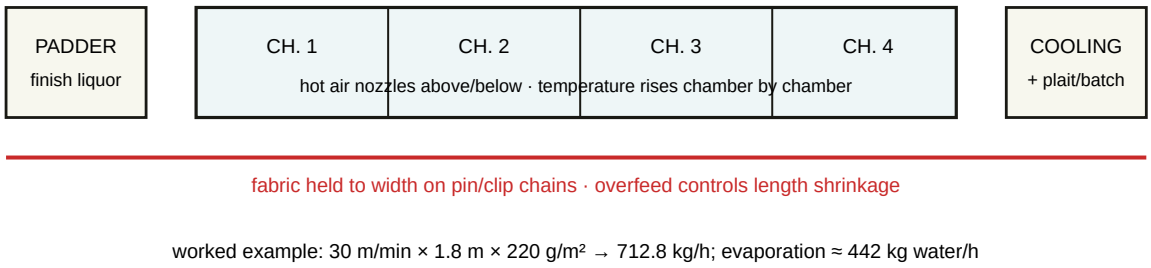


Fig. 1 — Stenter (schematic): padder applies the finish; pin/clip chains carry the fabric at set width through heated chambers that dry, heat-set and cure; overfeed at entry controls residual shrinkage.

The fabric is gripped at both selvedges by pins or clips on two endless chains that can diverge or converge to set width. Hot air is blown from nozzle boxes above and below in a series of chambers with independent temperatures — gentle first chambers (drying without baking migrating finish to the surface), hotter later chambers for heat-setting synthetics or curing resin finishes (typically 150–180 °C for cure). **Overfeed** — feeding fabric faster than the chain speed — lets warp yarns contract and pre-

shrinks the length; underfeed stretches. Exit moisture is monitored continuously (radio-frequency sensors) because the last percent of moisture is the most expensive to remove and the first cause of uneven cure.

WORKED EXAMPLE 1 — STENTER THROUGHPUT AND DRYING LOAD

Speed 30 m/min; width 1.8 m; fabric 220 g/m² (0.22 kg/m²).

Throughput = 30 × 60 × 1.8 × 0.22 = **712.8 kg/h** of dry fabric.

Fabric enters at 70% moisture on dry weight and leaves at 8%: water evaporated = 712.8 × (0.70 − 0.08) = **441.9 ≈ 442 kg/h**.

The evaporation figure — not the fabric figure — sizes the burners and exhaust fans; a stenter is a dryer that also sets dimensions.

3. Dimensional control machines

- **Sanforizer (compressive shrinkage range):** woven fabric is pressed by a heated cylinder against a thick rubber belt that contracts as it leaves the pressure zone, mechanically compacting the warp so residual shrinkage falls within the "Sanforized" tolerance. The standard answer for shirting and denim bottoms.
- **Compactor (for knits):** the knit is overfed onto a compacting zone of steam-heated felt belts (or shoe/blade units) that close up the courses; sets length and width of tubular and open-width knits and gives the soft final hand.
- **Relax dryer:** conveyor-belt dryers in which knits are carried without tension and tumbled by air, relaxing dyeing distortions before compacting.

4. Surface and appearance machines

MACHINE	EFFECT PRODUCED
Calender	Pressure (and heat) between bowls flattens yarns → smoothness, lustre, improved cover; embossing and friction (chintz) variants
Raising (brushing) machine	Wire-covered rollers tease out fibre ends → nap, warmth, softness (flannel, fleece)
Shearing / cropping	Rotating helical blades cut the raised nap to an even pile height
Sueding / emerising	Abrasive rollers give a fine peach-skin surface
Decatising	Steam and pressure set wool fabrics — lustre and dimensional stability (blowing/wrapping decatiser)

Tumbler / softening (air-jet type)	Mechanical softening and relaxation of garments and fabrics without chemicals

5. Matching machine to finish

Chemical finishing recipes (Topic 4) are only half the specification: the same add-on gives different results with different mechanical finishing. A resin-finished cotton cured on the stenter at 160 °C, then lightly calendered, reads crisp and smooth; the same fabric, tumbled after cure, reads soft. Finishing machinery selection is therefore part of recipe prediction: state machine, temperature profile, overfeed and moisture target beside the bath recipe, or the recipe cannot be reproduced.

6. Glossary — key terms

- **Overfeed:** delivering fabric faster than the chain/belt speed so it can contract in length.
- **Cure:** high-temperature fixation of a chemical finish in the stenter's hot chambers.
- **Compressive shrinkage:** mechanical pre-shrinkage by forcing the fabric to contract on a contracting belt.
- **Residual shrinkage:** the shrinkage remaining in finished fabric, measured by a standard wash test.

7. Key points

- The stenter dries, sets width, heat-sets and cures — its duty is computed in kg/h of fabric and kg/h of evaporated water.
- Dimensional stability is finished mechanically: sanforizer for wovens, compactor and relax drying for knits.
- A finishing specification is bath recipe plus machine settings; either alone is incomplete.

8. Summary

Finishing machinery converts wet, distorted cloth into a dimensionally stable product with a designed surface. The worked stenter — 712.8 kg/h of fabric, 442 kg/h of water — shows the scale of the flows the finishing engineer balances, and Topics 4 and 12 put those numbers to work in recipes and in plant design.

9. Practice

1. A stenter runs 25 m/min, 2.0 m wide, on 150 g/m² fabric entering at 60% moisture (on dry weight) and leaving at 7%. Calculate fabric throughput and evaporation rate.

2. Why are the first stenter chambers run cooler than the curing chambers when a resin finish is being dried and cured in one pass?
3. Match the machine to the requirement: (a) residual shrinkage of a woven shirting within ±1%; (b) even short pile on a raised blanket fabric; (c) width and length setting of cotton single-jersey.

Answer key

1. Throughput = 25 × 60 × 2.0 × 0.150 = 450 kg/h. Evaporation = 450 × (0.60 − 0.07) = 238.5 kg/h.
2. If wet fabric meets full cure temperature at once, water flashes off and finish migrates to the surfaces (uneven add-on, harsh hand, poor wash durability of the effect) before crosslinking. A rising profile dries gently first, then cures the uniformly distributed finish.
3. (a) Sanforizer; (b) shearing/cropping machine after raising; (c) compactor (with relax drying before it).
